

**Identifying Root Causes of Global Temperature Change Using Multi-Source Data
Integration and Regression Feature Ranking**

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CMPSC 445: Applied Machine Learning in Data Science

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March 17, 2026

Description of the Project

In this project, the goal was to create a regression-based model to find the root causes of global temperature change. Through the collection of data from multiple reputable sources, the project merges these datasets together to create a cleaned, merged, and concise dataset that can be used to train a linear regression model. Once this linear regression model is trained, the features are tested and ranked to determine the most influential factors in predicting global temperature change. From here, the data is plotted with 200+ plots to visualize the time-series, influential factors, and feature ranking.

Data Collection

To collect the data in this project, there were four main sources to obtain the data. First, I visited the NOAA Global Monitoring Laboratory website to obtain individual CSV files for the atmospheric greenhouse gases (CO₂, CH₄, and N₂O) (Lan et al., 2026; Lan et al., 2026). Then, I visited NASA's GISS Surface Temperature Analysis to obtain the global land-ocean average temperatures by month and year as another CSV file (NASA Goddard Institute for Space Studies, 2026). From here, I visited the NCEI NOAA website to obtain the solar irradiance data, which was obtained as a NC file (NCEI, n.d.). Lastly, I visited the Our World in Data website to obtain a long-term CO₂ concentration dataset to compare with the other CO₂ measure (NOAA Global Monitoring Laboratory, 2026). They had slightly different data from the NOAA, so the use of multiple sources is helpful to provide more information and perspectives.

Once I had all the CSV and NC files ready, I used the python file "dataCol.py" to create a function to read in all the data, create individual Pandas DataFrames in the program, and perform

individual formatting of the DataFrames. These DataFrames were returned as a list, and they were passed for data preprocessing.

Data Preprocessing

To prepare the data, using the Pandas library, I merged all the DataFrames together, and I dropped 2 columns that were included in the dataset that were the same across the entire set. From here, I formatted the column names, and I removed any missing values. This was a complex process as some of the datasets represented missing values different (with strings or null). Once these were filtered out, I replaced the empty values with the mean of the column. I did this as there was a long stretch of years that were pulled from different datasets, so there were many missing values for certain environmental measures. The use of the mean allows for the model to have access to a larger dataset and avoiding the process of shrinking the dataset to a significantly smaller size.

Then, I removed duplicated years by grouping. Grouping in Pandas allows for duplicated rows in certain columns to compress and merge into one instance of the given year (Aptovska & Gemini, 2026). For example, hypothetically, if there were three instances of the year 2026 where the first instance has the Annual CH₄ defined, but the second and third instances have Annual CO₂ and N₂O defined, using the group function in Pandas allows for all three instances to become one instances with all the attributes defined (Aptovska & Gemini, 2026).

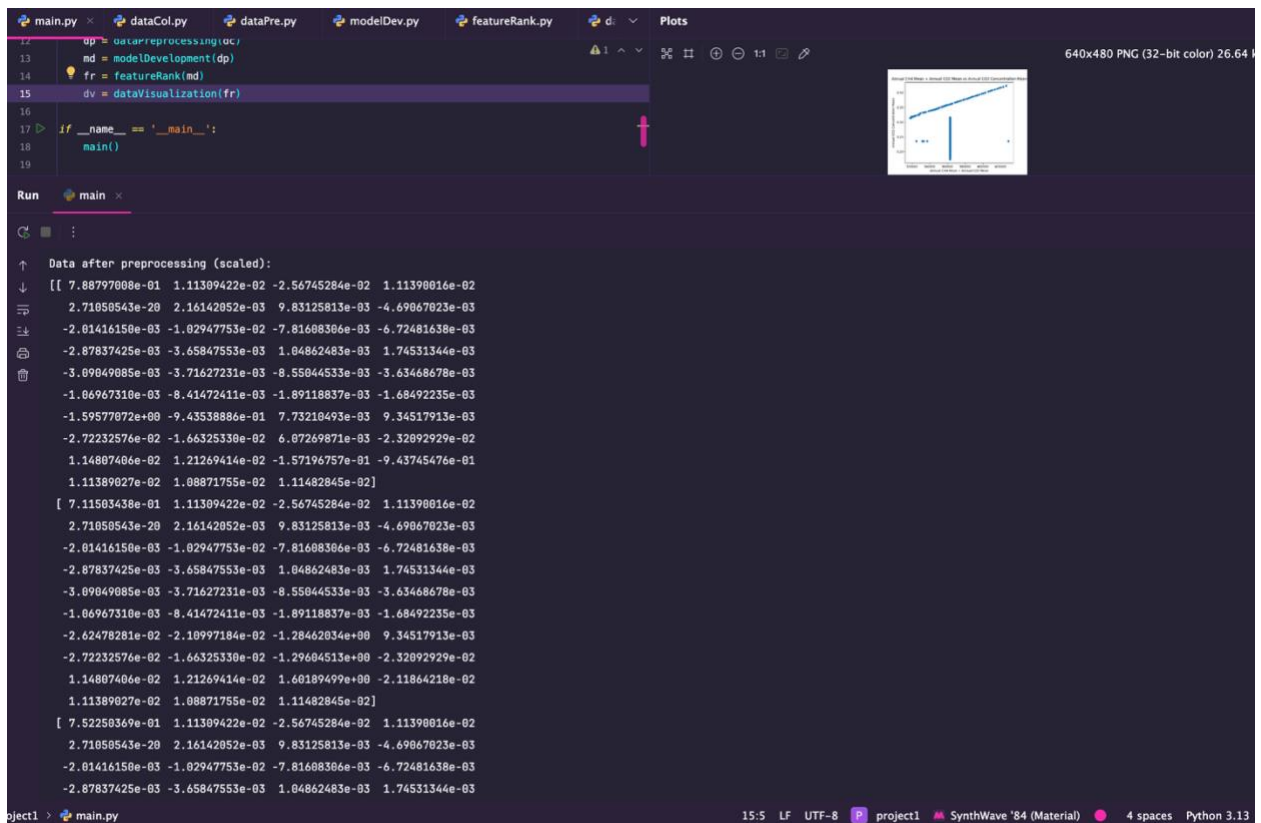
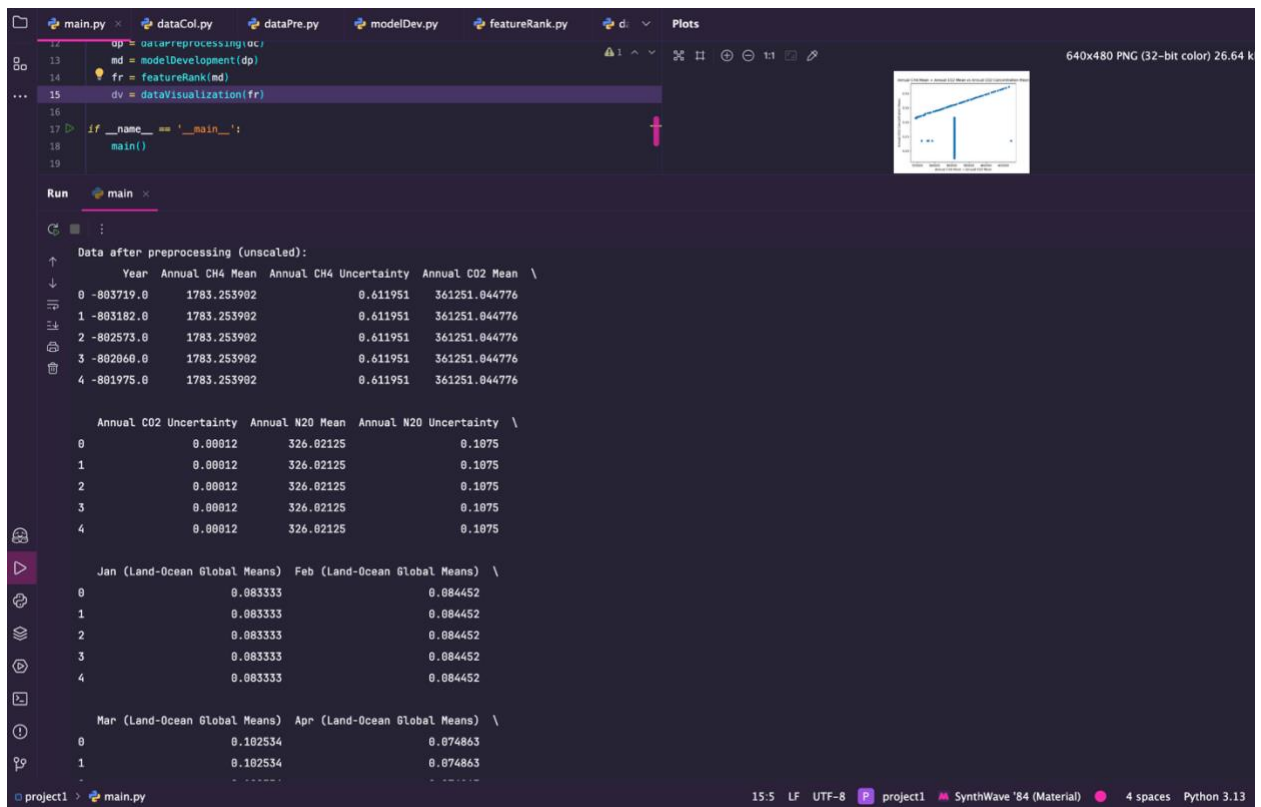
From here, I moved to unit conversion. The greenhouse gases were measured differently from each other, so I converted the CO₂ measurements to match the others. The CO₂ (from both sources) was measured in micromoles/moles and ppm, and the other greenhouse gases and their uncertainties were measured in nanomoles/moles and ppb. I converted the CO₂ by multiplying

by 1000 to convert. The other columns (such as temperature anomalies and total solar irradiance) are measured in their own formats, and all the temperature anomalies (monthly, seasonal, and yearly) were obtained by the same source. According to the NASA website, it is a “Land-Ocean Temperature Index” (NASA Goddard Institute for Space Studies, 2026). According to the NC file from NCEI, the total solar irradiance is documented as daily, monthly, and annual averages in W per m², which is the “total, spectrally integrated energy input to the top of the Earth's atmosphere, at a standard distance of one Astronomical Unit from the Sun” (NCEI, n.d.). I downloaded the yearly measure since all the other features are in yearly average measure.

Then, I engineered some more features. I created ratio features that showed the division between the different atmospheric gases. I tested dividing in both ways (larger divided by smaller and smaller divided by larger). It seems that many of the other attributes' values were less than one as well, and when the values were a ratio less than one, three of the engineered features made the top twenty features whereas only two made the top twenty when dividing by larger by smaller. This occurrence may have happened since there is more similarity across the non-engineered and engineered features when they are less than one. I chose to add both the ratio represented as a value larger than one and less than one to the dataset, and it seems that the ratios less than one were added to the top twenty features over the ones greater than one. However, the arrangement of the top 25 features also changed once these were added (further discussed in the feature ranking section). I also created addition features where I added the greenhouse gas concentrations together since they were in the same format measurement. Finally, once the data was cleaned and preprocessed with the features engineered, I moved the modified and merged data into its own CSV file to provide reusability.

Model Development

In this project, I chose to use a linear regression model to predict the target variable, temperature. From here, I used the developed dataset to create x (independent variables; predictors) and y (dependent variable; target). Then, I developed the train/test split for x and y using the scikit-learn library, creating a split of 70% training to 30% testing. Then, I used a standard scaler for the data since some of the features hold varying values that are significantly far apart. After the scaling process, I created the base linear regression model (including all the attributes), and I tested how well it performed with a mean squared error (MSE) score and R^2 score. The model performed quite well even without feature ranking. The MSE was around $4.4e^{-06}$, meaning there was a very low measure of error, and the R^2 score was 0.9996, a nearly perfect R^2 score. To improve further, I moved to feature ranking to not only develop a better linear regression model, but to also determine the root-cause features that predict the temperature anomaly best. The following figures show snippets of the output, showing parts of the dataset before and after scaling and the final output data. All 200+ generated plots can be seen by running the main.py file.



```

12 dp = dataPreprocessing(dp)
13 md = modelDevelopment(dp)
14 fr = featureRank(md)
15 dv = dataVisualization(fr)
16
17 if __name__ == '__main__':
18     main()
19
Run main
1.11389027e-02 1.08871755e-02 1.11482845e-02]]
MSE without feature ranking: 4.4818961546216396e-06
R^2 score without feature ranking: 0.9996491829988251
Support of feature: [False True False True False False False True True False False False
True True True False False False True True True True True True
False True True True True True True True True True False True
True False True]
Feature ranking: [14 1 13 1 15 6 9 1 1 8 7 10 1 1 1 2 3 4 1 1 1 1 1 1
12 1 1 1 1 1 1 1 1 1 1 1 1 1 5 1]
Feature names: ['Year', 'Annual CH4 Mean', 'Annual CH4 Uncertainty', 'Annual CO2 Mean', 'Annual CO2 Uncertainty', 'Annual N2O Mean', 'Annual N2O Uncertainty', 'Jan (Land-Ocean Global Means)', 'Feb (Land-Ocean Global Means)', 'Mar (Land-Ocean Global Means)', 'Apr (Land-Ocean Global Means)', 'May (Land-Ocean Global Means)', 'Jun (Land-Ocean Global Means)', 'Jul (Land-Ocean Global Means)', 'Aug (Land-Ocean Global Means)', 'Sep (Land-Ocean Global Means)', 'Oct (Land-Ocean Global Means)', 'Nov (Land-Ocean Global Means)', 'Dec (Land-Ocean Global Means)']
Top 25 features: ['Annual CH4 Mean', 'Annual CO2 Mean', 'Jan (Land-Ocean Global Means)', 'Feb (Land-Ocean Global Means)', 'Jun (Land-Ocean Global Means)', 'Jul (Land-Ocean Global Means)', 'Aug (Land-Ocean Global Means)', 'Sep (Land-Ocean Global Means)', 'Oct (Land-Ocean Global Means)', 'Nov (Land-Ocean Global Means)', 'Dec (Land-Ocean Global Means)', 'Annual CH4 Uncertainty', 'Annual CO2 Uncertainty', 'Annual N2O Uncertainty', 'Year', 'May (Land-Ocean Global Means)', 'Mar (Land-Ocean Global Means)', 'Apr (Land-Ocean Global Means)', 'Jun (Land-Ocean Global Means)', 'Jul (Land-Ocean Global Means)', 'Aug (Land-Ocean Global Means)', 'Sep (Land-Ocean Global Means)', 'Oct (Land-Ocean Global Means)', 'Nov (Land-Ocean Global Means)', 'Dec (Land-Ocean Global Means)']
MSE with feature ranking: 3.497969576916886e-06
R^2 score with feature ranking: 0.9997211594743663
MSE without top 25 features: 9.82659115250599e-05
R^2 score without top 25 features: 0.9921647362408025
Support of features (without top 25): [False False False False False False True True True True True True
False False]
Feature ranking (without top 25): [ 9 5 16 4 6 2 1 1 1 1 1 1 8 7 3]
Feature names (without top 25): ['Year', 'Annual CH4 Uncertainty', 'Annual CO2 Uncertainty', 'Annual N2O Mean', 'Annual N2O Uncertainty', 'Mar (Land-Ocean Global Means)', 'Apr (Land-Ocean Global Means)', 'May (Land-Ocean Global Means)', 'Jun (Land-Ocean Global Means)', 'Jul (Land-Ocean Global Means)', 'Aug (Land-Ocean Global Means)', 'Sep (Land-Ocean Global Means)', 'Oct (Land-Ocean Global Means)', 'Nov (Land-Ocean Global Means)', 'Dec (Land-Ocean Global Means)']
Top 5 features (without top 25): ['Apr (Land-Ocean Global Means)', 'May (Land-Ocean Global Means)', 'Sep (Land-Ocean Global Means)', 'Oct (Land-Ocean Global Means)', 'Nov (Land-Ocean Global Means)']
Process finished with exit code 0

```

Root-Cause Identification via Feature Ranking

For feature ranking, I completed two main tests. First, I created an RFE selector as my feature ranking method. Using this method, I tested the model with 15, 20, and 25 top features. It seems that there was improvement in the R^2 score only when selecting top 25 features, meaning that this is the ideal number of features for this feature selection/ranking. When I only engineered ratio features, it seems 20 features was showing an improvement in the R^2 score, but once I added more engineered features (added gases together), at 25 features is when the R^2 score would improve rather than decline. During this stage, I also compared the difference in the improvement and feature selection for the top 25 features when completing feature engineering. I tested both methods of ratios (smaller and larger than one), and I found that I should include both in the dataset and allow for the selector to decide between them.

In my top 25 features, I found that the selector gravitated towards the monthly and seasonal logs of the temperature anomaly to predict the yearly which is expected. Once I moved from including 15 to 20 features after the feature engineering process, I found that the selector was also selecting the annual greenhouse gases measure of CO₂, N₂O, and CH₄. Then, once I engineered more features and moved to 25 features, the engineered features and greenhouse gas measures were being included in the selector's choices as well. After feature ranking and selection, I retrained a linear regression model with only these top 25 features. I found that the R² score increased from 0.9996 to 0.9997 and the MSE score changed from $4.4e^{-6}$ to $3.5e^{-6}$. While this is not a huge difference and the model was performing well to begin with, it is important to recognize that feature ranking/selection can improve model performance even when the model is already performing quite well. In a scenario where the model did not have good initial performance, feature engineering, ranking, and selection would be essential steps to improve the model performance.

As an additional test, I tried to remove all the top 25 features to see the results that could be obtained if the best features were removed. The top 5 features without the main top 25 features were monthly logs of the temperature and the Annual CO₂ Concentration Mean (a second source's CO₂ measures). The MSE score was higher at $9.83e^{-5}$, and the R² score decreased to 0.9922. While these measures are still quite good, this goes to show that there is an effect on the model performance if the best features are not being used for training.

Visualization

To visualize the data, I created over 200 plots of the data. First, I started with creating simple line charts of each of the greenhouse gas measures against the year. I ensured to not include the data points that were the filled-in mean value because they would not show an

accurate representation of the data. While this imputed value is helpful in model training, it must be not included in data plotting so that the visualization is accurate to the actual representation of the data. Next, I plotted the MSE and R^2 scores that I found the three times they were measured. This line plot was not very interesting, but it shows that as MSE decreases R^2 increases. Then, I plotted a bar chart to represent each of the features ranking. Lastly, the bulk of the plots is plotting each of the top 25 features against each of the four greenhouse gas measurements both as a line plot and as a scatter plot.

Discussion and Conclusion

From the results found from the project, the most powerful predictors of the global annual temperature anomaly are the three greenhouse gas measurements from the NOAA Global Monitoring Laboratory (Lan et al., 2026; Lan et al., 2026), the monthly and seasonal temperature measurements from NASA's GISS Surface Temperature Analysis (Lan et al., 2026; Lan et al., 2026), and the engineered features of the greenhouse gas measurements from the NOAA Global Monitoring Laboratory (2026) and Our World in Data (NOAA Global Monitoring Laboratory, 2026). I believe this stems from the fact that as the years progress, temperature has increased and so have greenhouse gas emissions due to the manufacturing and automotive industries. Additionally, the monthly and seasonal temperature anomalies have increased throughout the years, so they are an indicator of how the global annual temperature anomaly will measure up to. I would argue these results align with climate science since global warming has been known to be a huge concern because greenhouse gases have on global warming. Based on these results and data, it is essential that safe practices are implemented regarding emissions to reduce greenhouse gases effects and overall reduce the effects of global warming.

During the data collection process, a challenge was ensuring that the correct data files were being downloaded and processed. Additionally, finding the total solar irradiance was a bit of a challenge as it was not listed directly on the NASA website. During the data merging process, it was a challenge to ensure that no data was being lost, the data was being properly merged, and the units were preserved. In this process, I merged the multiple frames together by keeping IDs on each of the dataframes columns to ensure no columns were merged before they were processed properly. During the data visualization process, one of the key challenges was ensuring that I was not plotting the mean values of each column. The year range for this dataset is very extensive since one of the datasets measures back to over 800,000 years ago, therefore, there were many values that were imputed. To create an accurate visualization, for certain columns, I chose to only plot values that were not the mean value, so the trend of the data is visible.

There are certain limitations to linear regression models because they casually infer that relationships and predictors all have linear relationships. Additionally, if features are strongly correlated, it can cause multicollinearity. It is crucial that complex data that is outside of the scope of linear models is modeled with more complex models to show the true relationship in the data.

Regarding data reliability, in this project, the data is quite reliable as it comes from national, reputable scientific organizations that are experts in data collection for this type of study. It is essential when collecting data that it is collected from multiple data sources to ensure that there is variety in the data in case there is miscalculations in any data. As for scientific accuracy, using data, such as the ones used in this project, that is from scientific organizations and laboratories is essential to ensure accuracy.

Regarding AI usage in this project, I used Gemini to suggest an idea for a group-by function during data preprocessing (Aptovska & Gemini, 2026). Other than this, the other AI that was used was Google search's AI Overview, and the results obtained from this were mainly syntactical and some suggestions, suggesting syntax fixes and functions in the Pandas library and Matplotlib library (Google search AI Overview, 2026).

Overall, this project led to good results with a high R^2 score and a low MSE, showing the importance of feature engineering and proper data collection and preprocessing. The machine learning pipeline for linear regression models has many steps, but they are all essential in producing a well-performing model.

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